

FINAL PLAN: June 24 - July 20 (27 Days)

This plan starts AFTER the Trilogy test (June 23). Pre-June-23 work (Nimblize report + Trilogy prep) is handled separately.

What Both Plans Agree On (The Non-Negotiables)

Both my plan and Claude's plan converge on these truths:

1. **Java + Backend + DB = your ONE path** (not two competing paths)
2. **TransitPulse is THE project** - finish it before starting anything new
3. **MongoDB for TransitPulse** (schemaless, beginner-friendly, no SQL needed to start)
4. **Daily DSA is non-negotiable** - Zift challenge + LeetCode streak
5. **Push to GitHub daily** - even imperfect code with comments
6. **Don't spread across 8 things** - serialize tasks by deadline
7. **One deployed live project > five half-finished ones**
8. **SQL/relational DBMS is still needed** for AKTU 5th sem - but AFTER TransitPulse

Where Plans Differed (And What I Picked)

Topic	My Plan	Claude's Plan	Final Decision
Database for TransitPulse	PostgreSQL/MySQL	MongoDB + Mongoose	MongoDB - faster to start, matches TransitPulse needs (real-time bus locations, TTL for stale data)
Deployment platform	AWS Free Tier	Render	Render first (simpler), then try AWS later as cloud learning
DBMS learning	Full SQL + relational DB theory	MongoDB only	MongoDB first (for TransitPulse), then SQL basics (for AKTU) in Phase C

Spring Boot timing	Phase 2 (June 23 onwards)	After July 10	After July 10 - agree with Claude, finish DB first
DSA scope	Full topic coverage	Focus on Topics 1+2, light on 3+4	Continue all 4 topics but prioritize Easy+Medium
DAA prep	2 full weeks	Minimal	1 week light intro - full prep during semester
C++ for DAA	Start in Phase 3	Minimum viable	Minimum viable - just syntax differences from C, learn during semester

The 3 Phases

Phase	Dates	Days	Primary Focus	Secondary
A	June 24 - July 6	13	MongoDB + TransitPulse DB	Java (GFG) + DSA
B	July 7 - 13	7	TransitPulse DB completion + Spring Boot start	DSA + Nimblize (if Phase 3 tasks come)
C	July 14 - 20	7	Spring Boot + Deploy to Render + DAA intro	SQL basics (AKTU prep) + portfolio

PHASE A: MongoDB + TransitPulse DB (June 24 - July 6)

What You'll Learn

- What is MongoDB, documents, collections (not tables/rows)
- MongoDB Atlas (free cloud hosting)
- Mongoose (MongoDB library for Node.js - schema definition)
- The 4 TransitPulse collections: Users, Routes, Buses, LiveLocations

TransitPulse Database Schema (The 4 Collections)

USERS

- _id
- name
- email
- password (hashed)
- role: "admin" | "driver" | "passenger"
- createdAt

ROUTES

- _id
- routeName: "Route 12A"
- stops: [{name, lat, lng, order}]
- isActive: true/false

BUSES

- _id
- busNumber: "UP32-1234"
- routeId (references Routes)
- driverId (references Users)
- capacity: 40
- status: "active" | "inactive"

LIVE_LOCATIONS

- _id
- busId (references Buses)
- lat, lng
- speed
- timestamp
- TTL index: auto-delete after 5 minutes (stale data cleanup)

Week 3: June 24-29

Day	Morning (2 hrs)	Afternoon (1.5 hrs)	Evening (1 hr)
Tue 24	Trilogy session + review test results	MongoDB 101: What are documents, collections, BSON. Watch a 30-min intro video	Zift daily + LC #383 Ransom Note (Hashing)
Wed 25	Set up MongoDB Atlas (free tier) + install	Java GFG: OOP - Classes, Objects, Inheritance	Zift daily + LC #349 Intersection of Two

	Mongoose + connect		Arrays
Thu 26	Write User.js Mongoose schema + test creating a user	Java GFG: Polymorphism, Abstraction, Interfaces	Zift daily + LC #347 Top K Frequent Elements
Fri 27	Write Route.js schema with embedded stops array	Java GFG: Collections - ArrayList, HashMap	Zift daily + LC #128 Longest Consecutive Sequence
Sat 28	Write Bus.js schema with references to Route and User	Java GFG: Collections - HashSet, LinkedList, Stack	Zift daily + LC #560 Subarray Sum Equals K
Sun 29	REST DAY	Light review of schemas written so far	-

Week 4: June 30 - July 6

Day	Morning (2 hrs)	Afternoon (1.5 hrs)	Evening (1 hr)
Mon 30	Write LiveLocation.js schema with TTL index (auto-delete)	Java GFG: Exception Handling + Strings	Zift daily + LC #238 Product of Array Except Self
Tue 1	Trilogy session	Write db.js connection file + test connection to Atlas	Zift daily + LC #48 Rotate Image
Wed 2	Seed dummy data: 1 route (3 stops), 2 buses, 3 users (admin/driver/passenger)	Java GFG: Multithreading basics	Zift daily + LC #50 Pow(x, n)
Thu 3	Verify seed data in Atlas UI + test queries (find all buses on a route)	Java GFG: File I/O	Zift daily + LC #73 Set Matrix Zeroes
Fri 4	Add indexes: email unique on Users, TTL on LiveLocations, routeld on Buses	Nimblize tasks (if any new assignment)	Zift daily + LC #189 Rotate Array

Sat 5	Test everything end-to-end: create user -> create route -> assign bus -> send location -> verify TTL deletes	Push all TransitPulse DB code to GitHub with proper README	Zift daily + LC #371 Sum of Two Integers (Bits)
Sun 6	REST DAY	Review + catch up on anything incomplete	-

Phase A Checklist

- ☐ MongoDB Atlas account set up
 - ☐ 4 Mongoose schemas written (User, Route, Bus, LiveLocation)
 - ☐ db.js connection file working
 - ☐ Seed data inserted and visible in Atlas
 - ☐ TTL index working (stale locations auto-deleted)
 - ☐ Java: OOP, Collections, Strings, Exception Handling covered
 - ☐ 12 LeetCode problems solved
 - ☐ Daily Zift challenges done + pushed to GitHub
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PHASE B: DB Completion + Spring Boot (July 7-13)

What You'll Do

- Finalize TransitPulse DB (any remaining work)
- Start Spring Boot** (Telusko YouTube playlist)
- Build your first REST API
- Continue DSA

Week 5: July 7-13

Day	Morning (2 hrs)	Afternoon (1.5 hrs)	Evening (1 hr)
Mon 7	TransitPulse DB: Final testing + write README documentation	Spring Boot: What is Spring Boot + project setup (Telusko)	Zift daily + LC #200 Number of Islands
Tue 8	Trilogy session	Spring Boot: Your first REST controller + GET endpoint	Zift daily + LC #70 Climbing Stairs

Wed 9	Spring Boot: POST/PUT/DELETE endpoints + request body	Spring Boot: Connect to a database (H2 or MongoDB)	Zift daily + LC #198 House Robber
Thu 10	TransitPulse DB: SUBMIT (deadline). Final verification in Atlas.	Spring Boot: JPA/Mongoose integration + CRUD operations	Zift daily + LC #322 Coin Change
Fri 11	Spring Boot: Service layer + proper project structure	Spring Boot: Build a mini "Product API" (CRUD)	Zift daily + LC #55 Jump Game
Sat 12	Spring Boot: Error handling + validation	Push Spring Boot project to GitHub	Zift daily + LC #62 Unique Paths
Sun 13	REST DAY	Review Spring Boot concepts	-

Phase B Checklist

- ☐ TransitPulse DB submitted by July 10
 - ☐ Spring Boot: Can create REST APIs with CRUD operations
 - ☐ Spring Boot mini project on GitHub
 - ☐ 6 LeetCode problems solved
 - ☐ Daily Zift challenges maintained
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PHASE C: Deploy + DAA + Portfolio (July 14-20)

What You'll Do

- Deploy TransitPulse (or Spring Boot project) to **Render** (free)
- Learn SQL basics for AKTU 5th sem DBMS subject
- Light DAA intro (Abdul Bari - just Unit I: asymptotic notations + sorting)
- Clean up GitHub portfolio for semester

Week 6: July 14-20

Day	Morning (2 hrs)	Afternoon (1.5 hrs)	Evening (1 hr)
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Mon 14	Deploy to Render: sign up + connect GitHub repo + deploy	SQL basics: CREATE TABLE, INSERT, SELECT (W3Schools)	Zift daily + LC #104 Max Depth Binary Tree
Tue 15	Trilogy session	SQL: WHERE, ORDER BY, JOINS	Zift daily + LC #1143 Longest Common Subsequence
Wed 16	Fix deployment issues + verify live URL works	DAA: Asymptotic Notations - Big-O, Omega, Theta (Abdul Bari)	Zift daily + LC #15 3Sum
Thu 17	DAA: Recurrence Relations + Master Theorem (Abdul Bari)	SQL: GROUP BY, Aggregate functions	Zift daily + LC #33 Search Rotated Sorted Array
Fri 18	DAA: Sorting algorithms overview - Merge Sort, Quick Sort theory	Portfolio: Update GitHub profile README	Zift daily + LC #78 Subsets
Sat 19	Portfolio: Add project screenshots, clean up repos, write descriptions	Revision: Java concepts + MongoDB + Spring Boot	Zift daily + LC #46 Permutations
Sun 20	BUFFER DAY - catch up on anything incomplete	Prepare for semester start	-

Phase C Checklist

- ☐ One project deployed live on Render with working URL
- ☐ SQL basics covered (enough for AKTU DBMS start)
- ☐ DAA Unit I introduced (asymptotic notations + sorting theory)
- ☐ GitHub portfolio polished with README + project descriptions
- ☐ 6 LeetCode problems solved
- ☐ Ready for 5th semester

Full LeetCode Tracker (Post June 23): 24 Problems

Phase A: Hashing, Arrays, Math, Bits (12 problems)

#	LC ID	Problem	Difficulty	Topic
1	#383	Ransom Note	Easy	Hashing
2	#349	Intersection of Two Arrays	Easy	Hashing
3	#347	Top K Frequent Elements	Medium	Hashing
4	#128	Longest Consecutive Sequence	Medium	Hashing
5	#560	Subarray Sum Equals K	Medium	Hashing/Prefix Sum
6	#238	Product of Array Except Self	Medium	Adhoc
7	#48	Rotate Image	Medium	Adhoc/Matrix
8	#50	Pow(x, n)	Medium	Maths
9	#73	Set Matrix Zeroes	Medium	Adhoc/Matrix
10	#189	Rotate Array	Medium	Adhoc
11	#371	Sum of Two Integers	Medium	Bits

Phase B: Graphs, DP, Greedy (6 problems)

#	LC ID	Problem	Difficulty	Topic
12	#200	Number of Islands	Medium	Graphs
13	#70	Climbing Stairs	Easy	DP
14	#198	House Robber	Medium	DP
15	#322	Coin Change	Medium	DP
16	#55	Jump Game	Medium	Greedy
17	#62	Unique Paths	Medium	DP

Phase C: Mixed + Placement Staples (6 problems)

#	LC ID	Problem	Difficulty	Topic
18	#104	Max Depth Binary Tree	Easy	Trees

19	#1143	Longest Common Subsequence	Medium	DP
20	#15	3Sum	Medium	Two Pointer
21	#33	Search Rotated Sorted Array	Medium	Binary Search
22	#78	Subsets	Medium	Backtracking
23	#46	Permutations	Medium	Backtracking

Daily Routine (Final)

Good Day (4-5 hrs):

- Morning (2 hrs): Primary task (MongoDB/Spring Boot/DAA)
- Afternoon (1.5 hrs): Secondary task (Java GFG/SQL/Nimblize)
- Evening (1 hr): Zift daily challenge + 1 LeetCode + push to GitHub

Busy Day / Low Energy (1-1.5 hrs):

- Evening only: Zift challenge + push to GitHub
- Never zero. Even 30 minutes counts.

Tuesday (Trilogy Day):

- Trilogy session
- Evening: Zift + LeetCode + review what Trilogy covered

Sunday:

- Rest. Light review at most.
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Rules (Final)

1. **TransitPulse DB first. No new projects until July 10.**
 2. **MongoDB for TransitPulse, SQL for AKTU** - learn both, but MongoDB first.
 3. **Java is your primary language.** C++ is DAA-lab-only (minimal, during semester).
 4. **Push to GitHub daily.** Imperfect code with a comment > no code.
 5. **Zift daily challenge = your non-negotiable minimum.** Even on bad days.
 6. **Deploy ONE thing to Render by July 20.** That live URL is your placement weapon.
 7. **Don't panic about Jan 2027.** You have 7 months. This plan covers 1 month. The remaining 6 months (5th semester) will deepen everything.
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What You'll Have By July 20

What	Proof
MongoDB skills	TransitPulse DB (4 collections, indexes, TTL, seed data)
Java proficiency	GFG topics covered + 24 LeetCode in Java
Spring Boot basics	Mini REST API project on GitHub
DSA foundation	Zift daily + LeetCode profile + Trilogy training
SQL basics	Enough for AKTU DBMS semester start
DAA awareness	Unit I intro (asymptotic notations, sorting theory)
Live deployed project	TransitPulse or Spring Boot app on Render
GitHub portfolio	Daily commits, clean repos, project READMEs
Nimblize credibility	2 professional reports submitted
Cloud understanding	Nimblize cloud report + Render deployment experience

That is a strong foundation for a 3rd-year student heading into placement season.

Quick Links

Resource	Link
GFG Java	geeksforgeeks.org/java
MongoDB Docs	docs.mongodb.com/manual
MongoDB Atlas	mongodb.com/cloud/atlas (free tier)
Mongoose Docs	mongoosejs.com/docs
Telusko Spring Boot	YouTube @Telusko - Spring Boot playlist
Abdul Bari Algorithms	YouTube - Abdul Bari Algorithms playlist
LeetCode	leetcode.com
W3Schools SQL	w3schools.com/sql

Render (deploy)	render.com
Draw.io (diagrams)	app.diagrams.net